

02: Need of JDBC

In the world of software development, data is at the core of most applications. Applications are built to process, manage, and store data, ensuring this data is stored and retrieved efficiently.

There are two types of storage:

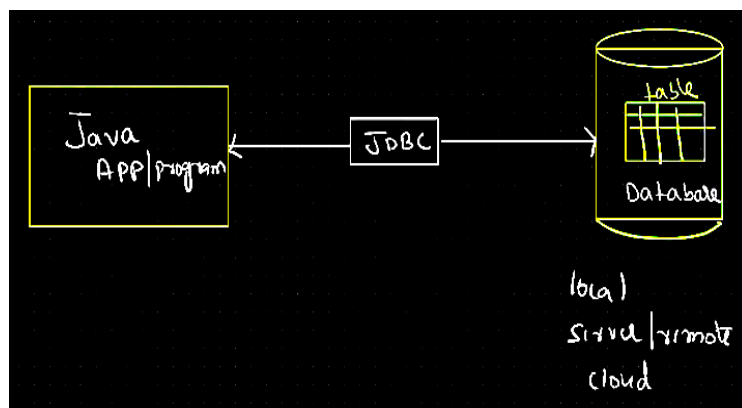
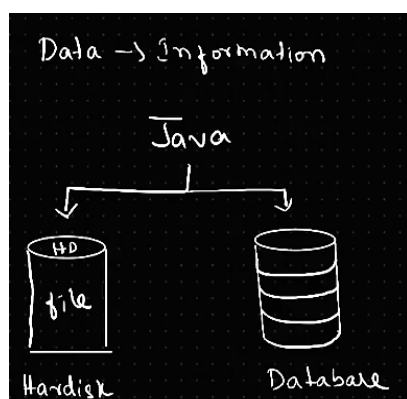
1. **Temporary Storage:** Applications and programs execute in *RAM* (Random Access Memory), which temporarily stores data. This data will be lost once the application stops or the system shuts down.
2. **Permanent Storage:** For long-term storage of data, it must be stored on a permanent medium like a hard disk or a database. This ensures data persists beyond the lifecycle of the application.

👉 File handling in Java:

- **File Storage:** If you need to store data permanently from a Java application, one option is to write the data to a file on the hard disk.

Java provides the *java.io.* package*, which includes classes and methods for IO (Input/Output) operations and file handling.

- **Persistency:** By using the *java.io.* package*, data can be persisted in a file on the hard disk, enabling it to be retrieved and reused later.



👉 Need for JDBC:

- **Database Storage:** Databases provide a more organized, secure, and scalable solution for data storage.

- **Connecting Java Applications to Databases:** JDBC enables Java programs to connect to various types of databases (local, remote, or cloud-based) and perform operations like querying, updating, and managing data.
- **Flexibility:** With JDBC, developers can interact with different databases (such as MySQL, Oracle, PostgreSQL, etc.) uniformly without worrying about the underlying database-specific details.